

Quantum teleportation formulas

The equations from the protocol pages: unknown state, shared Bell pair, Bell measurement, and Bob's corrections. The relation-lock note is included at the end.

What you start with

Alice holds an unknown qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Alice and Bob already share a Bell pair, usually

$$|\Phi^+\rangle_{A'B} = (|00\rangle + |11\rangle) / \sqrt{2}$$

Total state. Three qubits: payload A, Alice's half A', Bob's half B.

$$|\psi\rangle_A |\Phi^+\rangle_{A'B}$$

The protocol

1. Alice does a Bell measurement on A and A'. She measures in the Bell basis. That measurement has four equally likely outcomes.

$$|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle) / \sqrt{2}$$

$$|\Psi^\pm\rangle = (|01\rangle \pm |10\rangle) / \sqrt{2}$$

2. She sends the two classical bits that name which of the four outcomes she got. Entanglement is consumed. The particle is not moved, and the state is not sent as a signal.
3. Bob applies the matching Pauli to his half B. After that correction, B holds the original unknown state.

Bob's state before the correction

$$|\Phi^+\rangle \rightarrow |\psi\rangle_B = \alpha|0\rangle + \beta|1\rangle \quad (I)$$

$$|\Phi^-\rangle \rightarrow \alpha|0\rangle - \beta|1\rangle \quad (Z)$$

$$|\Psi^+\rangle \rightarrow \alpha|1\rangle + \beta|0\rangle \quad (X)$$

$$|\Psi^-\rangle \rightarrow \alpha|1\rangle - \beta|0\rangle \quad (ZX)$$

The symbol in parentheses is the operator Bob applies. ZX means X first, then Z, which restores $\alpha|0\rangle + \beta|1\rangle$.

Joint state written in the Bell basis

Expanding the three-qubit state so Alice's two qubits sit in the Bell basis:

$$|\psi\rangle_A |\Phi^+\rangle_{A'B} = (1/2) [$$

$$|\Phi^+\rangle (\alpha|0\rangle + \beta|1\rangle) + |\Phi^-\rangle (\alpha|0\rangle - \beta|1\rangle)$$

$$+ |\Psi^+\rangle (\alpha|1\rangle + \beta|0\rangle) + |\Psi^-\rangle (\alpha|1\rangle - \beta|0\rangle)]$$

Each Bell ket is on A A'. The state in parentheses is on B. The factor 1/2 makes the four outcomes equally likely.

The relation lock

Tandem correlation is not a pointer to a place. It is a lock between two records. Space stays the stage those records sit on.

The usable move for the quantum-spatial approximation: treat the lock as a relation, then ask what geometry or relocated state that relation can stand in for — after a classical channel is added, if you actually want to use it.